

Blue Ghost Mission 1 Mission Retrospective. B. M. Tackett, J. A. Marlin, and W. J. Coogan, Firefly Aerospace, Inc. (5900 Highway 183A Leander, TX 78641; benjamin.tackett@fireflyspace.com). © Firefly Aerospace, Inc.

Introduction: On March 2nd, 2025, the Blue Ghost Mission 1 spacecraft completed a soft and upright landing at Mare Crisium on the Moon. Part of NASA's Commercial Lunar Payload Services, the spacecraft carried ten NASA payloads which all completed their scientific objectives. The mission ended as planned on March 16th. Here we provide an overview of the mission and lessons learned.



Vehicle: Blue Ghost leveraged Firefly's composites and propulsion expertise to create a light-weight structure supporting a metric ton of propellant. This propellant fraction provided the delta-V needed to deliver the ten payloads to the lunar surface. Opting for a storable bipropellant, the vehicle was able to take an indirect path to the Moon, proving out all key systems prior to critical maneuvers such as lunar orbit insertion. With 400 watts of power, up to 125 kbps omnidirectional downlink over S-band, and 10 Mbps downlink over directional X-band, the indirect route afforded payloads additional data opportunities beyond the two weeks of surface operations. The lander also hosts a number of Firefly mechanisms. These include two two-axis gimbals and a surface access arm supporting five of the payload missions. Firefly performed assembly and integration of the vehicle in their Texas cleanroom, with integrated testing conducted at the Jet Propulsion Laboratory.

Launch and Transit: Blue Ghost Mission 1 launched on a SpaceX Falcon 9 from Cape Canaveral, Florida, on January 15th, 2025. It spent 25 days in Earth orbit performing system checkouts, phasing maneuvers, and science operations. Four days after its successful trans-lunar injection burn, the spacecraft performed a lunar orbit insertion maneuver to capture into an

elliptical lunar orbit. After 16 days of continued payload science operations, vision navigation checkouts, and orbital maneuvers and phasing, operators commanded Blue Ghost to perform Descent Orbit Insertion.

Landing: Once in its Descent Orbit, Blue Ghost engaged its vision navigation system to determine its location and attitude and began Powered Descent. Utilizing D'Souza guidance, the vehicle autonomously used both its main engine and eight Spectre thrusters to shed orbital velocity while navigating to its target landing site. Once over the landing site, the main engine disengaged, and the remainder of descent was performed with Spectre thrusters only. As the vehicle descended, two hazard avoidance maneuvers were autonomously commanded by the vision navigation system. Ground contact was sensed by three of the four footpad sensors, and successful soft and upright landing was verified using IMU data. LRO imagery verified successful touchdown 58 meters from the targeted landing ellipse.

Lessons Learned: The team documented and dispositioned almost 60 anomalies during the mission. As an example, our landing location near a crater combined with regolith accumulation on radiating surfaces caused temperature exceedances over the course of the lunar day, requiring an expansion of our planned quiescent period and innovative strategies such as using our patch array antenna as a sunshade.

End Of Mission: As intended, Blue Ghost operated five hours after the end of the lunar day, sending back photos of the sunset and possible dust levitation, a phenomenon first reported by Apollo astronauts. Eventually, the batteries depleted, and the lander avionics and radios shut down. The mission was concluded as Blue Ghost was not designed to survive the cold lunar night.

Looking Forward: Firefly engineers are already implementing lessons learned for BGM2, which will carry the LuSEE Night radio telescope to the far side of the Moon in 2026. This mission, and the recently awarded BGM3, add transfer stages, orbital relays, and rovers to the architecture, expanding the national capability and toolset at the Moon, building hope for bright future of commercial lunar exploration and science.